AI MOBILE APP FOR EARLY IDENTIFICATION OF STUDENT MENTAL HEALTH VIA BEHAVIOR & SPEECH

Project Id: 25-26J-278

Project Proposal Report

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MULTIMODAL EMOTION DETECTION WITH RL-POWERED PERSONALIZED INTERVENTIONS FOR STUDENT MENTAL HEALTH PLATFORM

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1 DECLARATION

DECLARATION

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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2 ABSTRACT

This research introduces a personalized real-time intervention platform aimed at improving student mental well-being. It integrates multiple ways to detect emotions, monitor physiological signals, and use reinforcement learning (RL). The system constantly tracks students' emotional states through facial expressions, heart rate, and behavioral signals during learning or counseling sessions. This gives a clear picture of their feeling's moment to moment [2][3]. Unlike traditional systems that react only after stress or disengagement happens, this platform uses proactive intervention timing through emotion forecasting. Sequence models like Transformers or LSTMs can predict stress or negative emotional peaks 30 to 60 seconds ahead [5] [19]. This allows the RL agent to intervene before students face increased stress. A key feature of the system is Emotion-Informed RL Reward Shaping through Clinician Feedback. Here, doctors or counselors rate how effective interventions are during actual sessions. This approach allows the RL agent to improve its decision-making not just based on engagement or compliance data, but also on expert clinical opinions. This makes each intervention more relevant and impactful. The interventions are adaptable and include guided mindfulness exercises, short physical activities, and interactive cognitive nudges. The delivery is tailored for timing, method, and intensity based on what each student prefers.

The system measures effectiveness using various behavioral, physiological, and psychological metrics. These include engagement patterns, heart rate stabilization, ecological momentary assessment (EMA) of mood, and validated clinical scales like PHQ-9 [14] and GAD-7 [15]. The platform is built for scalable, privacy-friendly use through mobile and web applications. It features low-latency processing, on-device capabilities, and secure cloud integration. By merging emotion-aware RL, proactive stress forecasting, and clinician-guided policy adjustments, this research presents a unique operational system for adaptive mental health interventions [16]. It offers culturally sensitive and real-time support for university students in Sri Lanka.

Key Words – Personalized intervention, Forecasting, Reinforcement, PHQ, GAD

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6 LIST OF ABBREVIATIONS

Abbreviations	**Description**
CNN	Convolutional Neural Networks
NLP	Natural Language Processing
LSTM	Long Short-Term Memory
AI	Artificial Intelligence
RL	Reinforcement Learning
EMA	Ecological Momentary Assessment
PHQ-9	Patient Health Questionnaire-9 (clinical depression scale)
GAD-7	Generalized Anxiety Disorder-7 (clinical anxiety scale)

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Appendix 1: Gantt Chart

Appendix 2: Work Break Down Chart

8 INTRODUCTION

Student mental health is a growing issue worldwide. Stress, anxiety, and depression impact academic performance and overall well-being [7]. In Sri Lanka, many university students experience high stress levels. However, traditional counseling and basic digital tools often fall short. They frequently lack flexibility, timely responses, and personalized engagement. This situation shows the need for scalable systems that can support students' emotional and cognitive needs in real time.

Recent advancements in artificial intelligence and reinforcement learning (RL) allow for the creation of personalized interventions [11]. By using various ways to detect emotions such as facial expressions, heart rate, and behavior patterns an RL agent can provide interventions tailored to individual needs [1] [15]. A key innovation is proactive stress forecasting. Using sequence models like Transformers and LSTMs (Long Short Term Memory), the system can predict emotional changes in advance. This enables intervention before stress escalates. Including feedback from clinicians in RL reward shaping ensures that interventions are based on both engagement metrics and expert evaluations, improving safety and relevance [6].

The proposed platform offers adaptive interventions like mindfulness exercises, micro physical activities, and cognitive nudges [13]. It also continuously monitors outcomes through behavioral and physiological metrics. Designed to protect privacy, the system works on mobile and web platforms. It integrates fast pipelines, on-device processing, and secure cloud infrastructure. By combining emotion detection, proactive forecasting, RL personalization, and clinician input, this research aims to create a culturally sensitive, real-time platform to improve student mental health in Sri Lanka.

8.1 LITERATURE REVIEW

8.1.1 Background & Literature Survey

The mental well-being of university students is an increasing concern, with high levels of stress, anxiety, and depression reported worldwide [8]. Traditional interventions, like counseling or workshops, often lack personalization and timely delivery, which limits their effectiveness. Recent developments in AI systems now allow for real-time monitoring and support of student mental health through multimodal emotion detection. This method combines facial expressions, heart rate, and behavioral patterns to assess emotional states. Studies suggest that multimodal approaches improve prediction accuracy compared to single-mode systems, which leads to more responsive and personalized interventions [2]. Reinforcement learning (RL) has also been used to optimize the timing and content of these interventions[23]. However, most existing systems rely only on engagement metrics and reactive strategies, without incorporating clinician expertise or predictive modeling.

Previous work on AI mental health tools showed that facial emotion recognition and physiological monitoring can identify stress and disengagement during online learning or counseling sessions. Yet, RL-based personalization rarely includes feedback from humans, where clinicians provide real-time ratings of intervention effectiveness to refine the RL policy[20]. Additionally, most systems only take action after detecting stress. By using sequence models like LSTMs or Transformers to predict emotional trends, it becomes possible to implement proactive interventions that prevent escalation and improve outcomes [4]. Combining multimodal sensing, RL personalization, clinician feedback, and proactive timing, this proposed research aims to fill gaps in current methods and provides a new, culturally adapted framework for improving student mental well-being.

8.2 RESEARCH GAP

Although recent studies have improved AI-driven mental health support, some limitations still exist. Most current systems rely on reactive methods, detecting stress only after it has escalated. This reduces the effectiveness of support strategies. For instance, Zhang and Li (2024) proposed a Transformer-based framework for predicting adolescent mental health, but their work focused solely on prediction without including real-time intervention methods [9].

Another issue is the lack of clinician feedback in reinforcement learning (RL) models. While RL has been used to tailor mental health recommendations, current models mainly optimize based on engagement measures instead of therapy effectiveness[21]. This gap lowers the clinical reliability of these systems (Zhou et al., 2023) [10].

Finally, even though multimodal techniques, such as combining facial, voice, and behavioral cues, have boosted emotion recognition accuracy, very few systems incorporate physiological signals like heart rate into real-time adaptive frameworks for students[18]. Existing systems like EmoPulse emphasize multimodal assessment but do not utilize predictive modeling for proactive interventions (Kumar et al., 2024) [12].

This research aims to fill these gaps by creating a multimodal, clinician-in-the-loop reinforcement learning system with proactive stress forecasting. The goal is to provide timely, personalized, and clinically appropriate interventions for students' mental health.

8.3 RESEARCH PROBLEM

University students experience rising levels of stress, anxiety, and depression. These issues harm their academic performance, social interactions, and overall quality of life [7]. Traditional mental health support, such as counseling sessions and workshops, often fails to adapt in real-time or cater to individual needs. This limits their effectiveness in addressing unique emotional states. While AI-driven tools for mental health have started to appear, most systems depend on reactive strategies that use engagement metrics or after-the-fact assessments. They usually do not include clinician feedback or predictive modeling [1].

Current approaches based on reinforcement learning improve the timing and type of interventions but seldom use feedback from clinicians, who could assess the effectiveness of interventions during sessions. Moreover, most systems intervene only after they detect stress or disengagement. This approach misses the chance for proactive support that could stop negative emotions from escalating [4]. There is a significant gap in providing timely, personalized support that combines multimodal emotion detection, RL optimization, and clinician knowledge.

The main research questions this study addresses is, how can we create a culturally adapted, real-time mental health support system for university students? [22] This system should integrate multimodal emotion detection, RL-based personalization, clinician feedback, and proactive intervention timing to effectively enhance emotional well-being and engagement.

9 OBJECTIVES

9.1 Main Objective

The main goal of this research is to create a personalized, real-time mental health support system for university students. This system will use various methods to detect emotions, personalizes through reinforcement learning, and include feedback from clinicians. It combines facial expression analysis, heart rate monitoring, and behavioral cues. The aim is to provide timely and culturally sensitive interventions that improve student well-being, lower stress and anxiety, and encourage participation in learning or counseling sessions. This research tackles the lack of proactive, personalized mental health support and sets up a framework for scalable, real-time intervention systems.

9.2 Sub Objectives

- To design and implement multimodal emotion detection using facial expressions, heart rate, and behavioral patterns to assess students' emotional states.
- To develop a reinforcement learning agent that chooses the best types of interventions and their timing based on real-time emotional and behavioral data.
- To integrate clinician feedback, allowing professionals to rate how effective interventions are during sessions to improve the RL policy.
- To apply proactive intervention strategies using sequence models, such as LSTMs and Transformers, to predict emotional changes and prevent increased stress or anxiety.
- To evaluate the system's effectiveness through student engagement metrics, emotional improvements, and feedback from both students and clinicians.
- To provide dashboards and reporting tools for educators, counselors, and administrators to track emotional trends, intervention outcomes, and system performance.

10 METHODOLOGY

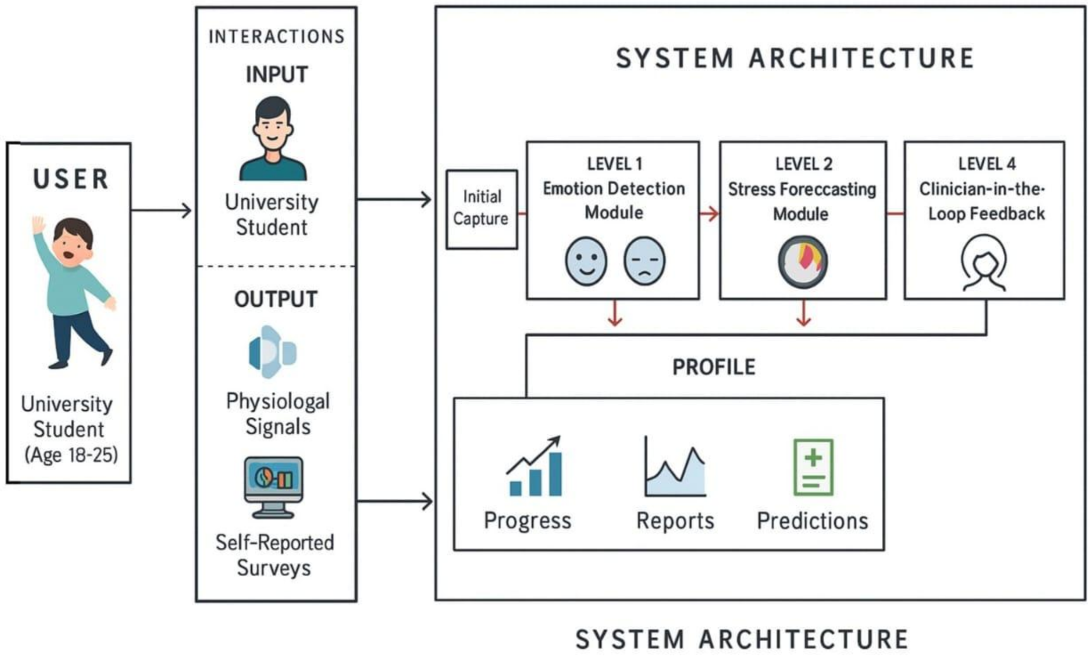

Figure 1: High Level Architecture Diagram

This study uses a design and development research method along with an experimental evaluation approach to create a culturally adapted, real-time mental health support platform for university students in Sri Lanka. The system employs multimodal emotion detection, reinforcement learning personalization, clinician feedback, and proactive stress forecasting to provide timely and effective interventions. The methodology includes system design, data collection, model development, implementation, and evaluation.

The system architecture has four main components. The multimodal emotion detection module captures facial expressions, behavior patterns, and physiological signals like heart rate to assess students' emotional states in real time. Facial recognition relies on convolutional neural networks, while wearable sensors offer continuous physiological monitoring. The proactive stress forecasting module uses sequence models such as

Long Short-Term Memory networks and Transformers to anticipate short-term emotional trends, enabling preemptive interventions before stress builds up.

The reinforcement learning personalization module customizes interventions based on each student's emotional responses and engagement patterns. The RL agent trains using a reward system influenced by engagement metrics and clinician feedback, optimizing the timing, type, and intensity of interventions. The clinician feedback module includes expert evaluations of intervention effectiveness, enhancing safety and relevance. Interventions like mindfulness exercises, micro physical activities, and cognitive nudges are offered through mobile and web platforms, with secure processing on devices and in the cloud to ensure privacy and quick responses.

Data collection involves university students aged 18 to 25 in Sri Lanka, all providing informed consent. The system gathers physiological data (heart rate), behavioral data (app usage, engagement duration), and facial recordings. Self-reported stress, anxiety, and mood surveys serve as ground truth labels. Ethical considerations include anonymization and compliance with data protection regulations.

Model development includes multiple layers. CNNs process facial expressions, while LSTM models analyze physiological signals. Sequence models like LSTM and Transformers predict short-term emotional trends for proactive intervention. The RL agent defines the state as the current emotional profile, actions as possible interventions, and rewards as a mix of engagement metrics and clinician-rated effectiveness, allowing ongoing learning of optimal intervention strategies.

The system is available on mobile (iOS/Android) platform, with secure cloud services for storage and computation. On-device processing guarantees low-latency emotion detection and intervention delivery. Adaptive pipelines manage multimodal input effectively. The platform offers a seamless, scalable, and culturally suited experience.

For evaluation, participants use the platform over 4 to 6 weeks. Baseline and follow-up measurements include stress reduction (through physiological data), engagement levels, and intervention response times. Qualitative metrics assess user satisfaction, perceived effectiveness, and clinician ratings. Comparative analyses evaluate the proactive RL-based system against reactive baselines lacking clinician feedback or forecasting. Statistical tests, including paired t-tests or ANOVA, assess improvements in emotional well-being and relationships between RL optimization, clinician feedback, and intervention success.

Expected outcomes include correct multimodal emotion recognition, timely personalized interventions that prevent stress escalation, improved engagement and emotional well-being, and a culturally adapted, scalable system suitable for Sri Lankan universities. Integrating multimodal sensing, RL personalization, clinician feedback, and proactive forecasting offers a new framework for real-time, adaptable mental health support for students.

11 PROJECT REQUIREMENTS

11.1 Functional Requirements

- **User Registration & Authentication**

Students can register and log in securely using email or university credentials. Password recovery is available.

- **Multimodal Emotion Detection**

Capture facial expressions, behavioral patterns, and physiological signals like heart rate, galvanic skin response, and sleep patterns to detect emotional states in real time. Use CNNs for facial analysis and LSTM models for physiological data.

- **Proactive Stress Forecasting**

Predict short-term emotional trends using LSTM and Transformer models. Trigger early interventions before stress levels increase.

- **RL-Based Personalization**

Change interventions based on individual emotional responses and engagement metrics. Continuously improve the type, timing, and intensity of interventions.

- **Clinician-in-the-Loop Feedback**

Clinicians give real-time ratings of how effective interventions are. Feedback is used in RL reward shaping to enhance personalization.

- **Intervention Delivery**

Provide mindfulness exercises, micro physical activities, and cognitive nudges through mobile and web platforms. Ensure timely, personalized, and low-latency delivery.

- **Data Collection & Management**

Collect and store physiological, behavioral, and facial data securely. Gather self-reported surveys on stress, anxiety, and mood. Ensure that data anonymization and compliance with protection regulations are maintained.

- **Evaluation & Analytics**

Track engagement, response times, stress reduction, and user satisfaction. Create reports for both students and clinicians.

11.2 Non-Functional Requirements

- **Performance**

Process multimodal inputs in real-time with minimal delay.

Achieve high accuracy in detecting emotions and predicting stress.

- **Scalability**

Support multiple users at once while handling growing amounts of data effectively.

- **Security & Privacy**

Encrypt data while it is being transmitted and when stored.

Follow Sri Lankan data protection laws.

Prevent unauthorized access.

- **Reliability**

Ensure system uptime with continuous monitoring.

Manage sensor or data failures smoothly.

- **Usability**

Provide an intuitive interface for students and clinicians.

Allow easy navigation and clear visualization of interventions and progress.

- **Maintainability**

Use a modular design that allows for easy updates.

Facilitate the integration of additional sensors, interventions, or analytics modules.

11.3 Expected Test Cases

- **Register a new student account**

Check that the account is created and a confirmation email is sent.

- **Login with valid credentials**

Verify successful login and access to the dashboard.

- **Login with invalid credentials**

Ensure access is denied and an appropriate error message is shown.

- **Detect facial emotion**

Confirm the system accurately identifies emotions from video or images in real-time.

- **Collect physiological data from wearable**

Verify accurate recording of heart rate, GSR, and sleep patterns.

- **Predict stress trend**

Check correct forecasting of stress levels and triggers for preemptive intervention.

- **Deliver personalized intervention**

Confirm the intervention is delivered according to RL policy through mobile or web.

- **Clinician feedback integration**

Ensure the RL agent updates intervention policies based on clinician ratings.

- **Data storage security**

Verify secure storage and prevention of unauthorized access.

- **System response under load**

Check that performance remains stable and latency stays low with multiple concurrent users.

- **Recovery from sensor failure**

Ensure the system continues monitoring and handles missing data smoothly.

12 BUDGET

Table 1 : Budget Table

Expenses	Requirement Cost (Rs.)	Justification
Travelling cost for surveys and awareness programs	15,000.00	To conduct initial data collection, distribute questionnaires, and raise awareness about burnout among students at universities.
Data Collection (standardized questionnaires, survey tools, digital forms)	10,000.00	Required for acquiring reliable datasets from students for training, testing, and validating the system.
Deployment cost (server hosting, backend services)	10,796.00 / month	To maintain system availability and reliability through cloud-based deployment.
Hosting cost in Play Store	8,076.00	One-time cost to publish the mobile application on the Google Play Store for student access.
Miscellaneous (printing, promotional activities, student awareness sessions)	5,000.00	For marketing, leaflets, and awareness campaigns to ensure successful adoption of the app.
Total Expenses	**48,872.00 + 10,796.00/month**	—

13 COMMERLIZATION

13.1 Target Audience

- University students aged 18–25 experiencing stress, anxiety, or mental health challenges
- University counseling centers and mental health support staff
- Parents or guardians of university students seeking mental health resources
- Educational institutions and NGOs promoting student well-being
- Government and private organizations focusing on youth mental health initiatives

13.2 Market Space

- Universities and higher education institutions in Sri Lanka and the South Asian region
- Mobile platforms for mental health apps (iOS App Store, Google Play Store)
- Mental health-focused organizations, counseling centers, and therapy programs
- Collaborations with research institutions, NGOs, and educational technology partners
- Potential integration with wearable device providers for physiological monitoring

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15 APPENDICES

Appendix 1: Work Break Down Structure

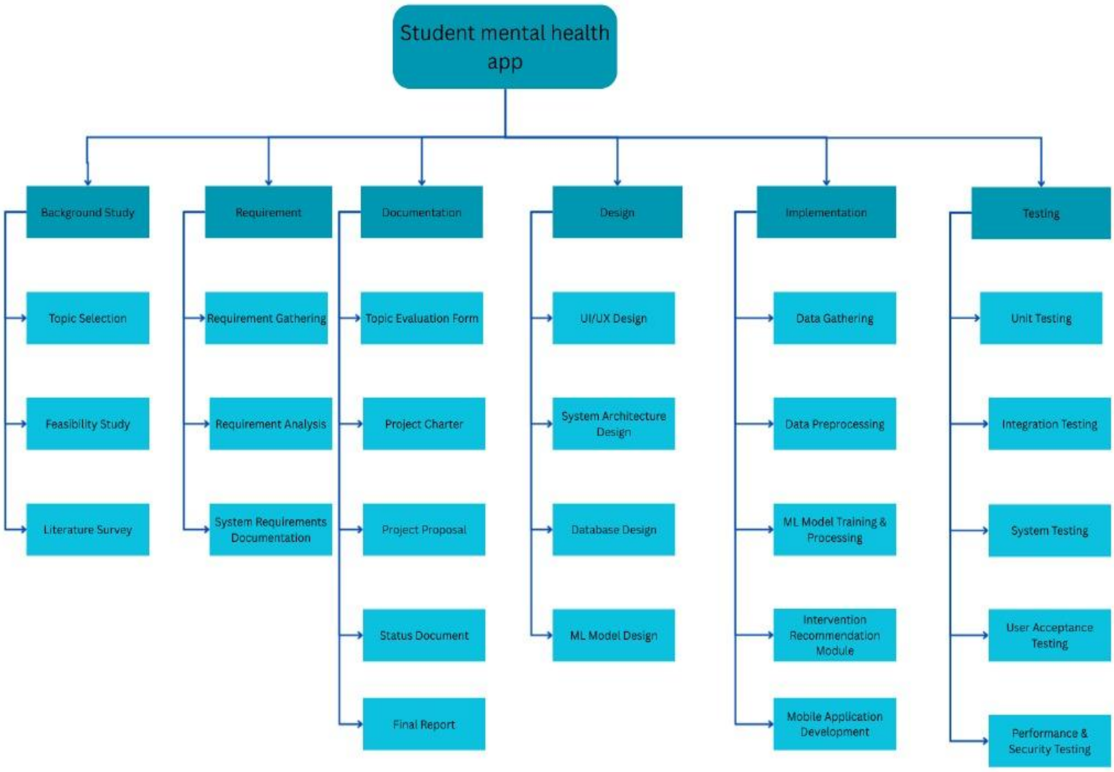

Figure 2 : Work Break Down Structure

Appendix 2: Gantt Chart

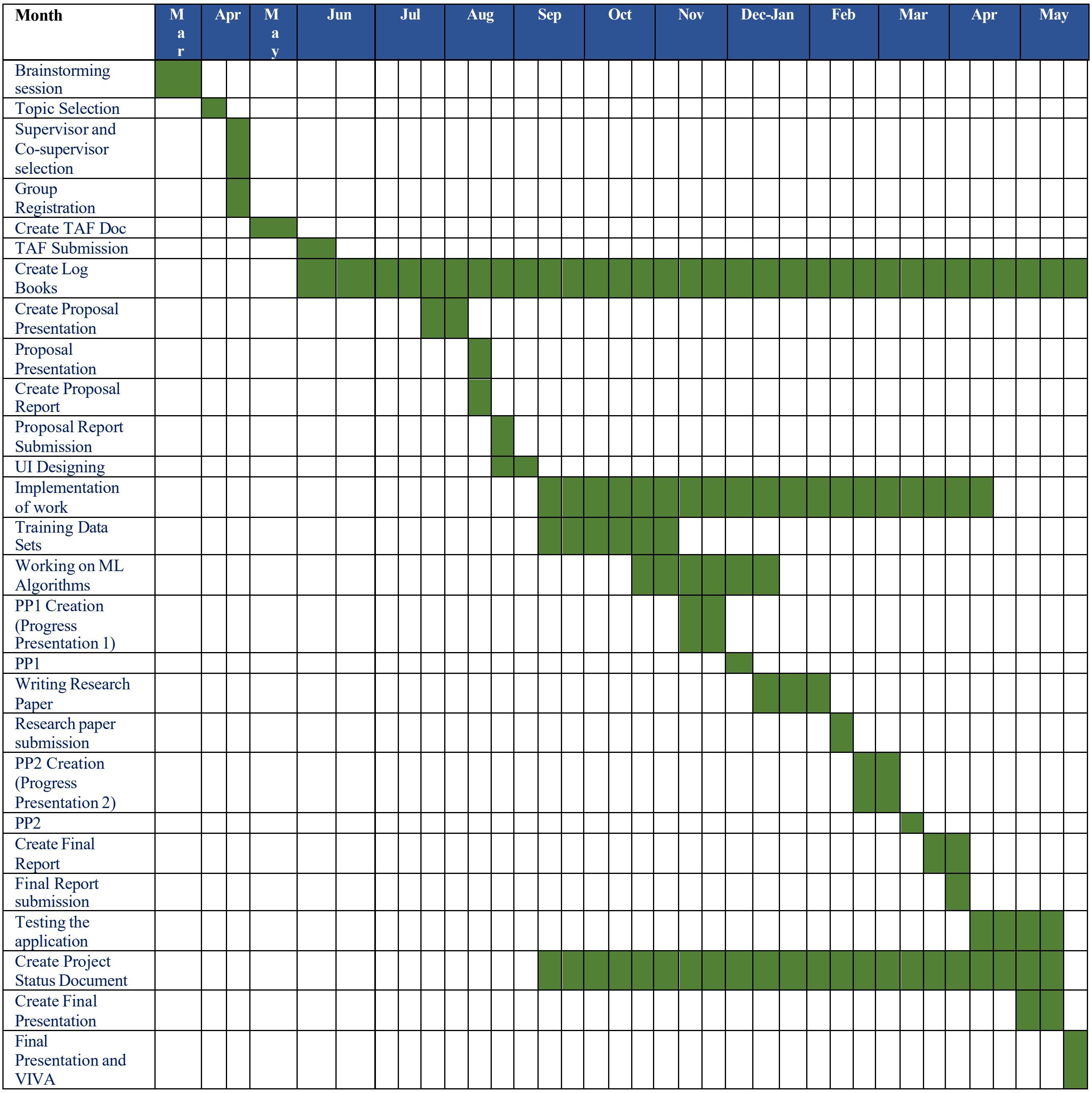

Figure 3 : Gantt Chart